The residual collector (large-prime limit) on prime curves

2026-10-26 · `research/ic_prime_residual_lp_20261026` · protocol frozen in `c77297bda/4aafbbffa` · six ecbench sessions, 1,536 records, all verified, audits OK · reference `rho.negation` · lead `LEAD-ICP-residual-partial-relations`.

Question

Does crediting residuals lower the counted cost of the prime `ic.pipeline` baseline? Crediting means storing every residual of a partial decomposition and turning collisions into relations: `ic_framework::residual`, the `|D| = r` limit of the large-prime variation. If it does, is the gain generic? The collector ran at base sizes 4–64 and at the plan's N_0, on six registered prime curves (20.6–32 bits), against the `ic plan` recipe and rho.

Result

$\log_2 r$	S plan	S res4	S rho	res4/plan w=0	w=0.1	w=1	res4/rho
20.59	10.59	4.51	1.19	0.426 [0.379, 0.479]	1.119 [0.983, 1.274]	3.886 [3.338, 4.525]	3.78
21.19	10.51	4.32	0.87	0.411 [0.333, 0.507]	1.104 [0.873, 1.397]	3.871 [2.972, 5.043]	4.94
24.00	13.15	3.95	0.98	0.300 [0.279, 0.324]	0.868 [0.801, 0.940]	3.067 [2.806, 3.351]	4.05
25.35	14.36	4.18	0.92	0.291 [0.251, 0.339]	0.866 [0.740, 1.013]	3.105 [2.600, 3.709]	4.54
28.00	20.40	3.72	0.93	0.182 [0.160, 0.207]	0.550 [0.482, 0.628]	1.949 [1.706, 2.228]	4.01
32.00	29.40	3.51	0.89	0.119 [0.104, 0.137]	0.367 [0.318, 0.424]	1.330 [1.137, 1.555]	3.95

Counted (w = 0), the 4-point collector is 2.3–8.4× cheaper than the plan; at 1 GAE per native operation it is 1.33–3.89× dearer. The order reverses inside the AGENTS.md §5a band at every size, so the two are not ranked. Exponents (slope of ln GAE on ln r): plan 0.634, collector 0.45–0.48 at every fixed base, rho 0.480. The collector has rho's exponent at about 4× rho's counted constant.

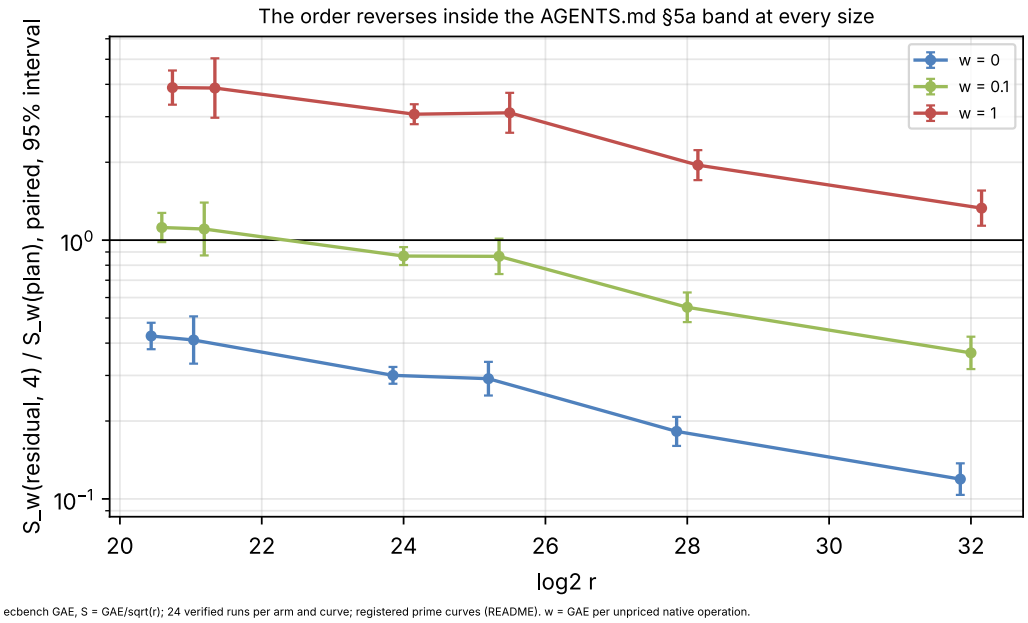


Figure 1: res4 against the plan at w = 0, 0.1 and 1, paired.

Curve-free null. The same collector and seeds on `Z/r` with a random-bijection key cost the same. The curve-free/curve ratio of mean group operations is 0.964–1.029 over p21, p24 and p28 × B = 4, 16 and 64, and all nine 95% intervals contain 1.

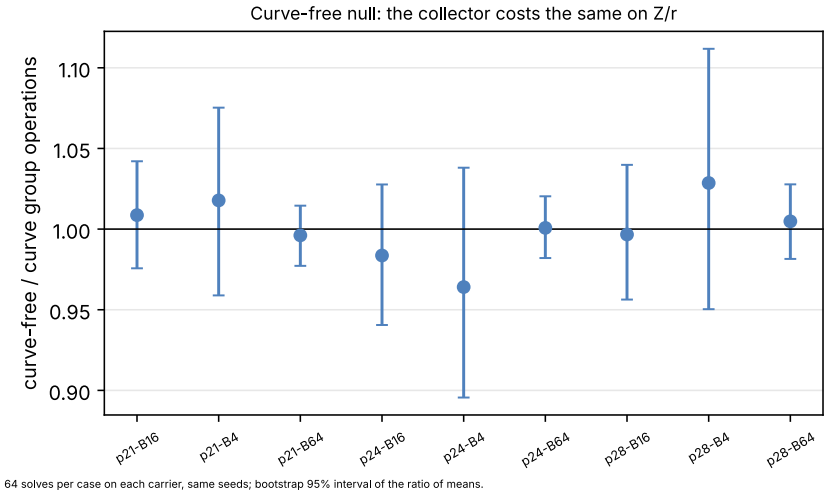


Figure 2: Curve-free / curve group operations per solve.

Decision

All six predictions held, with P1 holding only in the counted unit.

Classification. The result is generic collision search, band-dominated by the plan, so it moves no leaderboard row and no planner rule.

Proposed atlas disposition. `closed_known`: residual crediting on prime fields is the birthday bound. The w = 1 ratio falls with size and would cross 1 near $\log_2 r \approx 35$ if the trend held; that is an extrapolation, and both arms would still be at least 4× rho.

Deviations. No resume checkpoint could be uploaded (no token); the owner said to proceed, and every input and output is committed and pushed. The binary was built at `b561ebca8`, after the freeze; the change does not touch these arms.